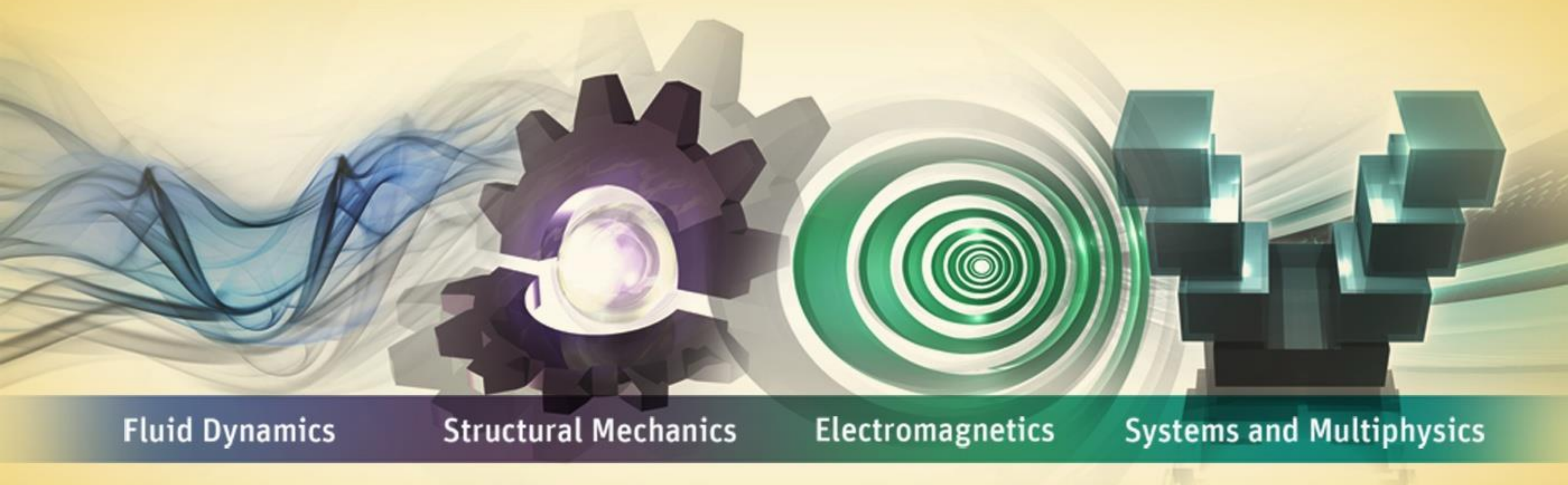
Realize Your Product Promise™

Workshop 5A:
Linear Eigenvalue Buckling Analysis

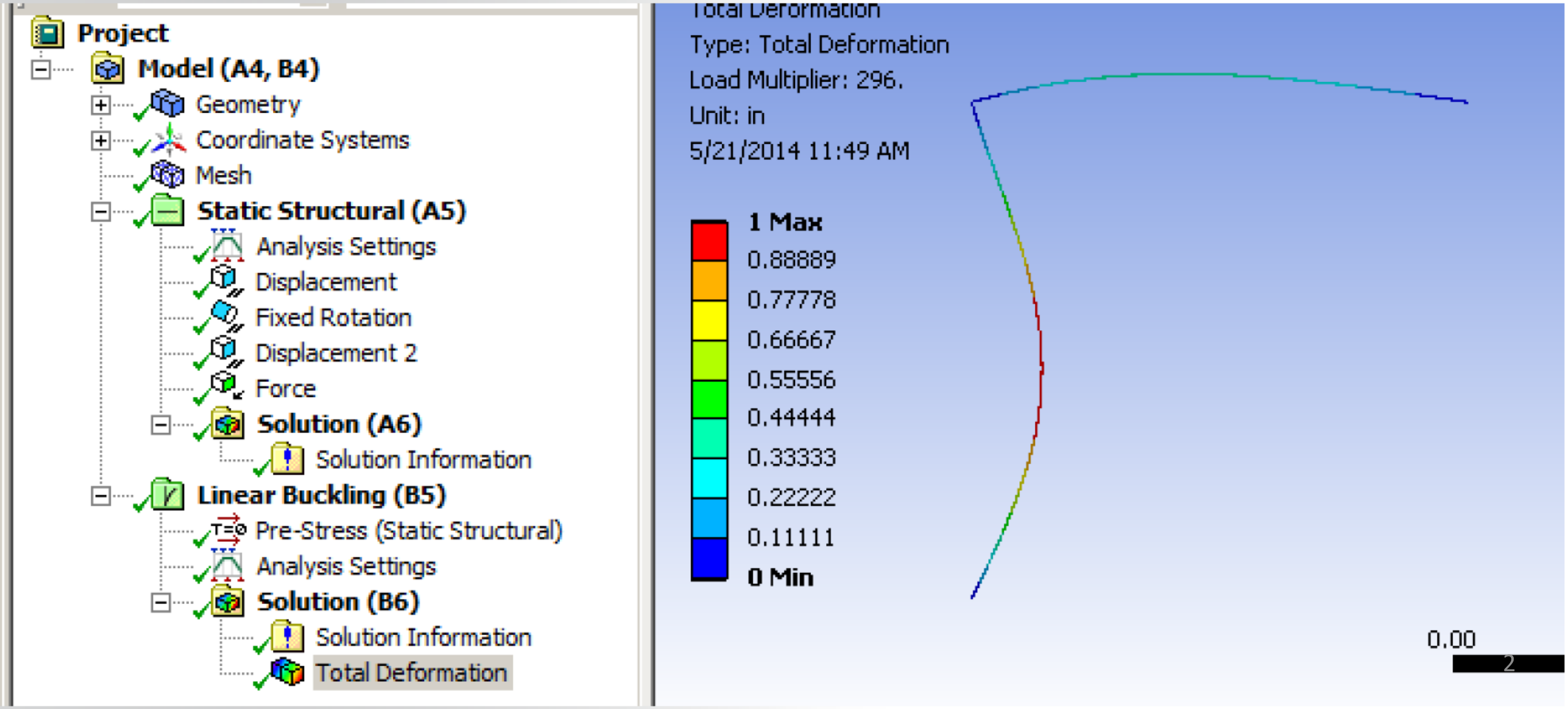


ANSYS Mechanical
Introduction to Structural Nonlinearities

Workshop 5A: Linear Eigenvalue Buckling

Goal

Use Eigenvalue Procedure to predict force necessary to produce buckling in a thin column.



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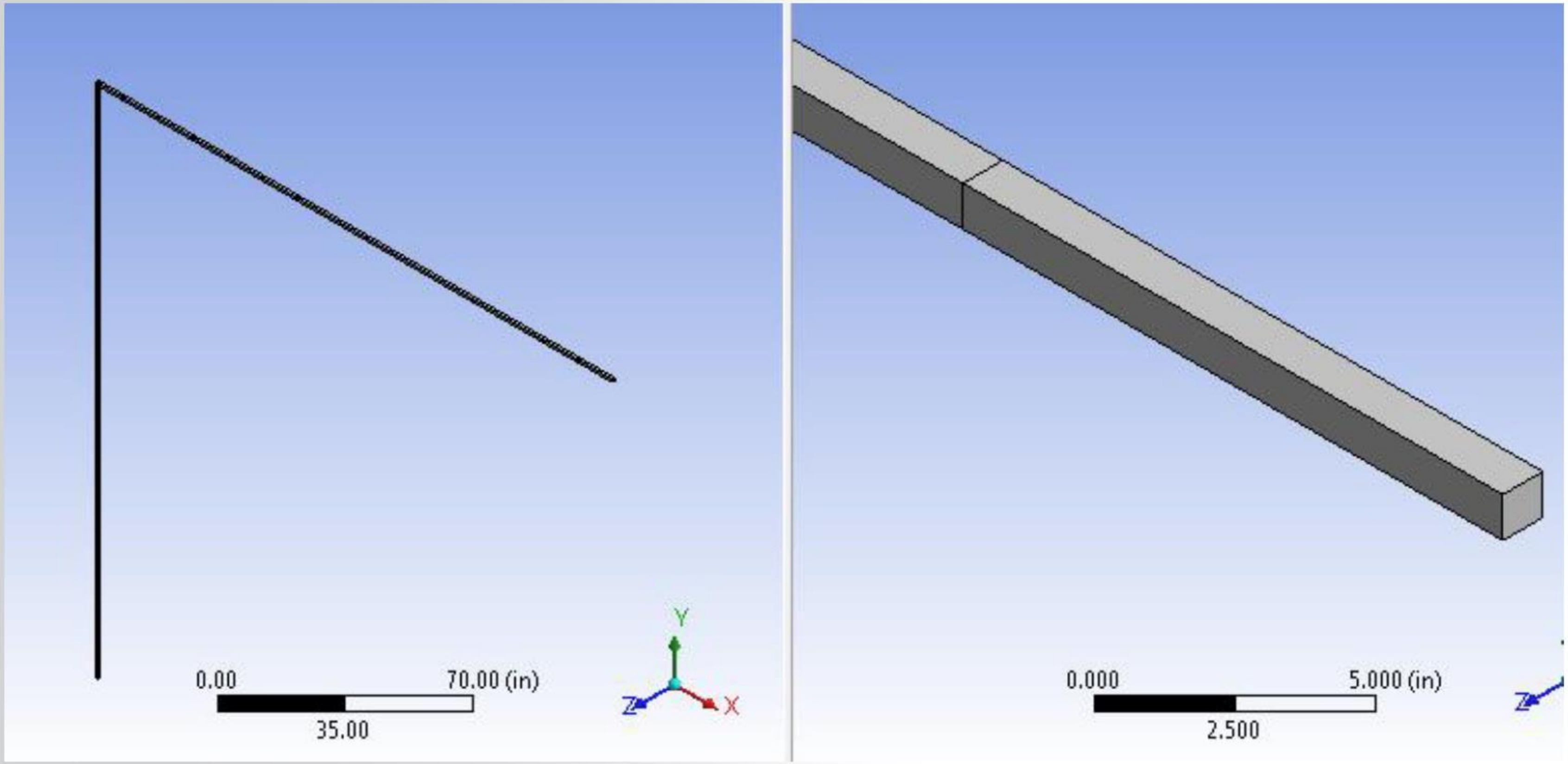
- <u>Model Description</u>
  - L shaped frame represented with 3D Line Bodies
  - Meshed with 3D Beam elements
  - Linear Elastic Material



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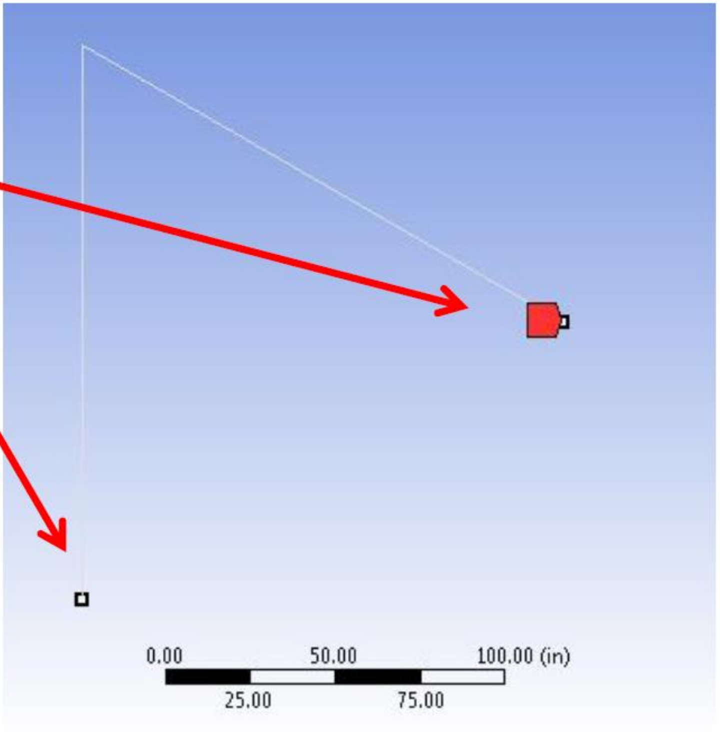
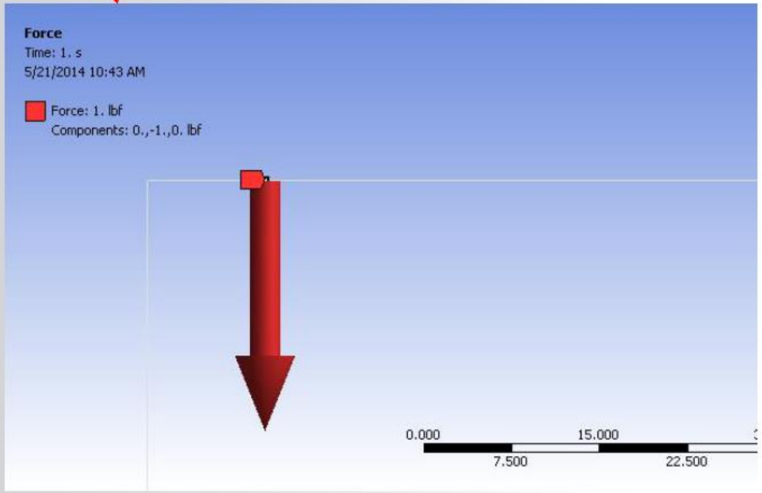
- Model Description (cont'd)
  - Horizontal beam fixed at far end.
  - Vertical column simply supported at bottom.
  - All nodes are constrained from movement in Z direction.
  - Compressive unit load (1 lb) applied 10 inches off center of vertical column.





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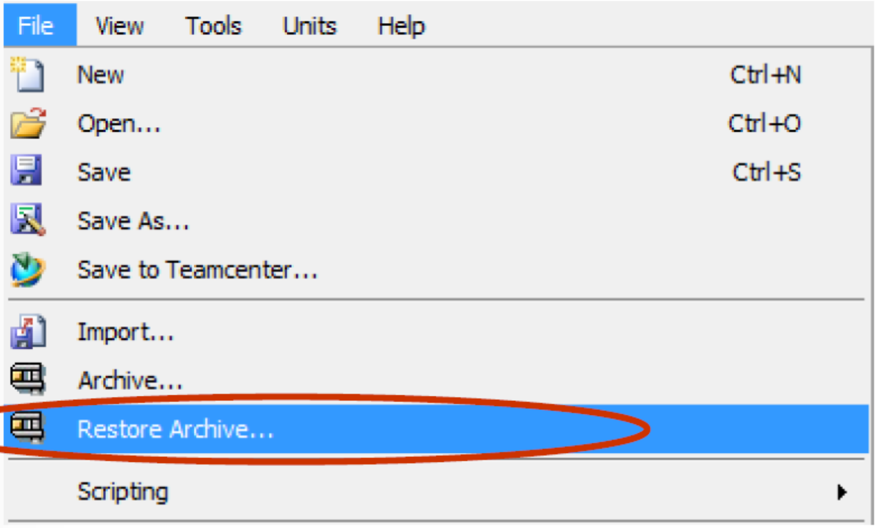
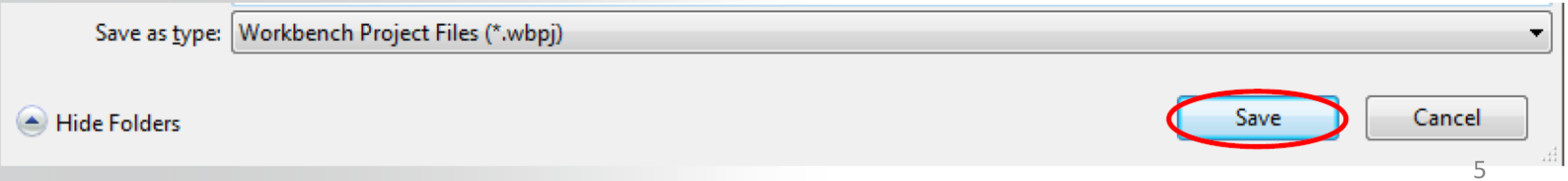
Steps to Follow:

Restore Archive… browse for file “SNL WS5A_buckling.wbpz”



Save as

- **File name: “WS5A_buckling”**
- **Save as type: Workbench Project Files (*.wbpj)**



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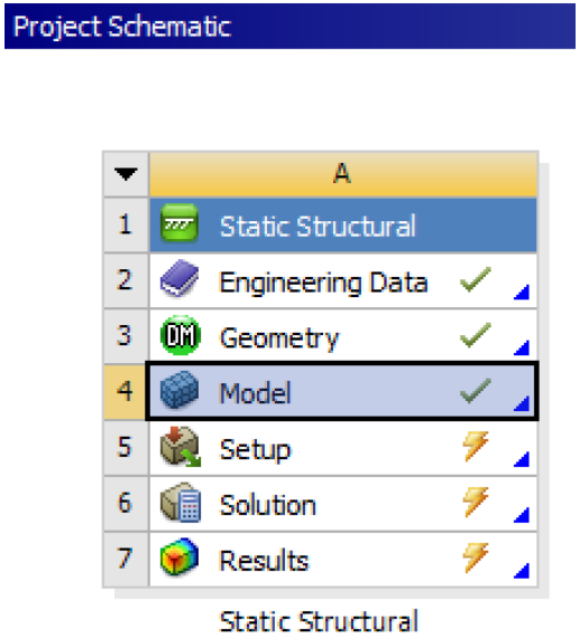
The project Schematic should look like the picture to the right.

- From this Schematic, you can see that the Engineering (material) Data and Geometry have already been defined (green check marks).
- It remains to set up and run the FE model in Mechanical
- Open the Engineering Data Cell (highlight and double click OR Right Mouse Button (RMB)>Edit)
- To see relevant dialog boxes, it might be necessary to go to Utility Menu > View..
  - Click on ‘Properties’ and ‘Outline’
- Verify that the units are in US Eng units (lb, in,…) system. If not, fix this by clicking on…
  - Utility Menu > Units > US Engineering(lb, in,…)



A	B	C
Property	Value	Unit
Density	0.2836	lb in^-3
Isotropic Secant Coefficient of Thermal Expansion		
Isotropic Elasticity		
Derive from	Young's Modulus an...	
Young's Modulus	1E+07	psi
Poisson's Ratio	0.3	
Bulk Modulus	8.3333E+06	psi
Shear Modulus	3.8462E+06	psi
Alternating Stress Mean Stress	Tabular	
Strain-Life Parameters		
Tensile Yield Strength	36259	psi
Compressive Yield Strength	36259	psi
Tensile Ultimate Strength	66717	psi
Compressive Ultimate Strength	0	psi

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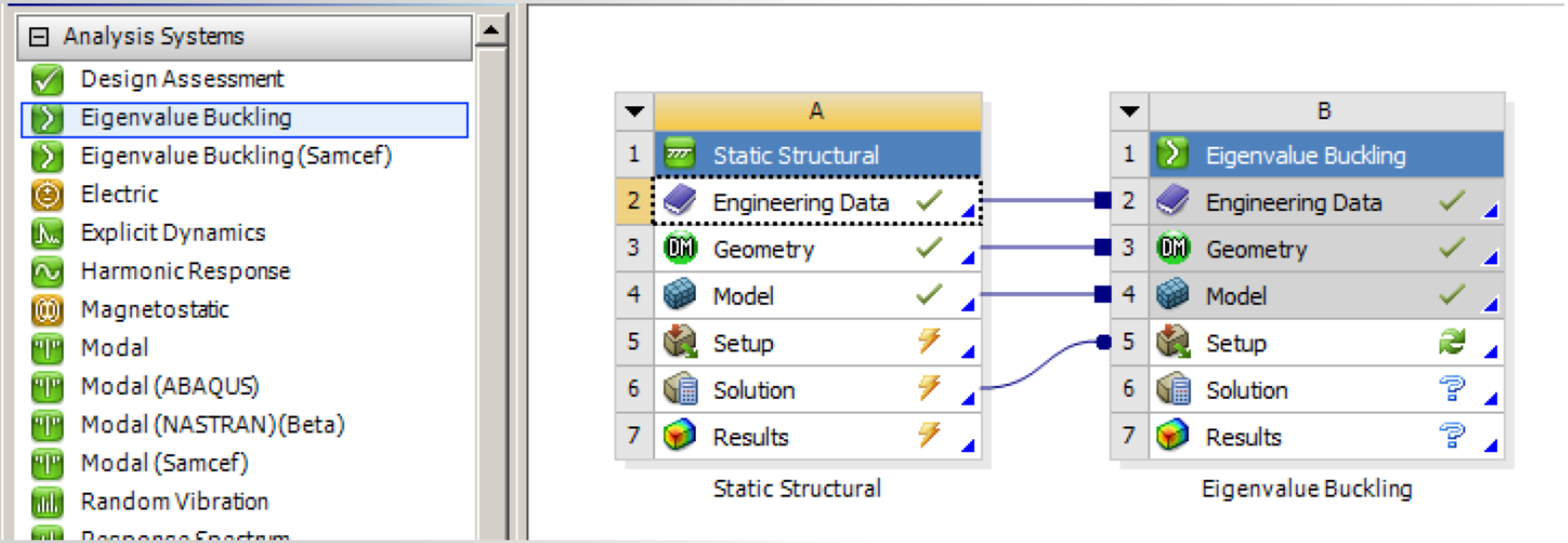
Return to Project Schematic

From the Analysis Systems Folder of the Toolbox, drag and drop an Eigenvalue Buckling Template onto the Solution Cell of the existing Static Structural Template and Refresh the Project.



Open Mechanical (Double click on the "Setup")

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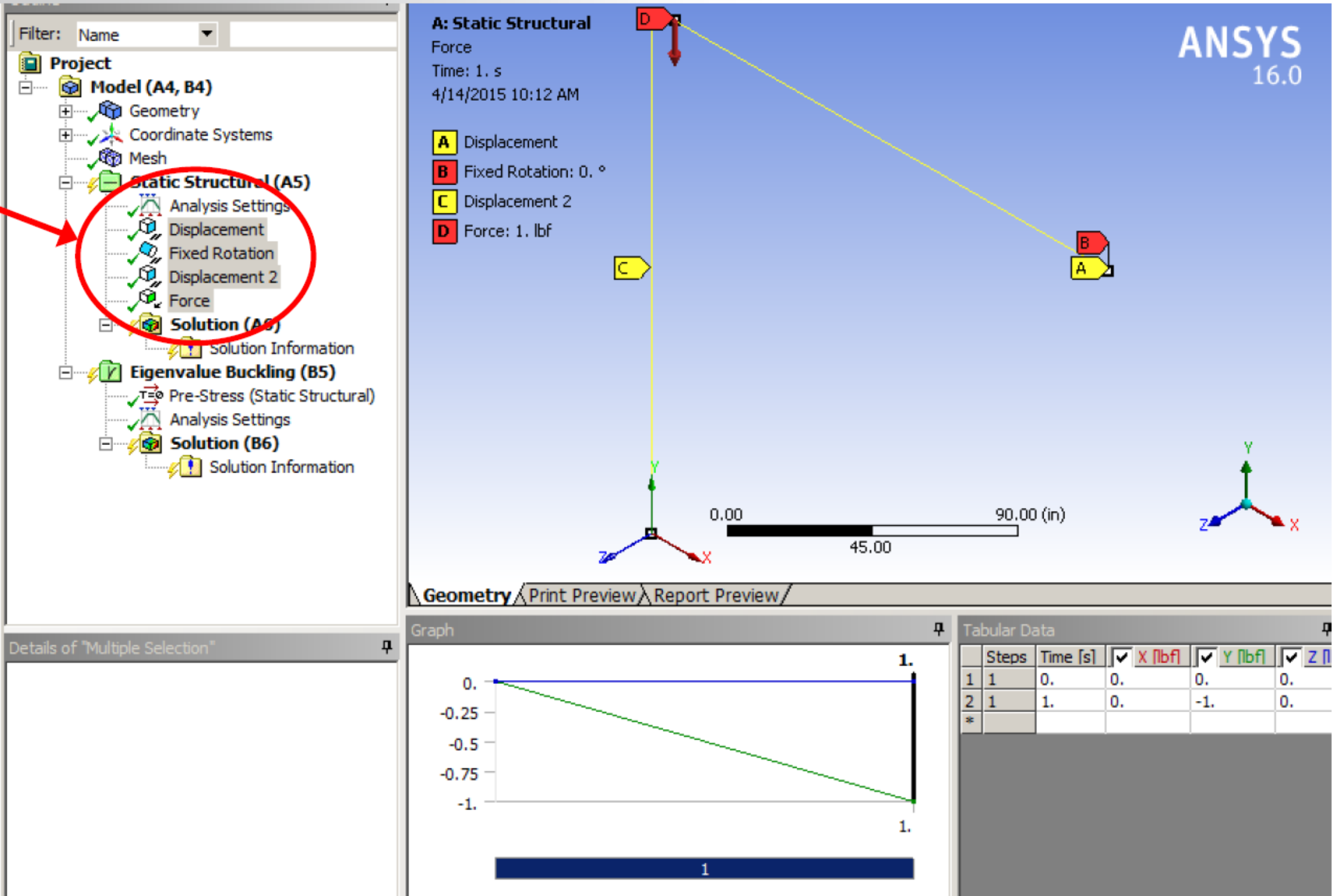
The frame model is already set up with the displacement boundary conditions and a unit force load.

- Highlight the BCs and Load to confirm that the model is properly supported and loaded.



Steps	Time [s]	X [lbf]	Y [lbf]	Z [
1	0.	0.	0.	0.
1	1.	0.	-1.	0.

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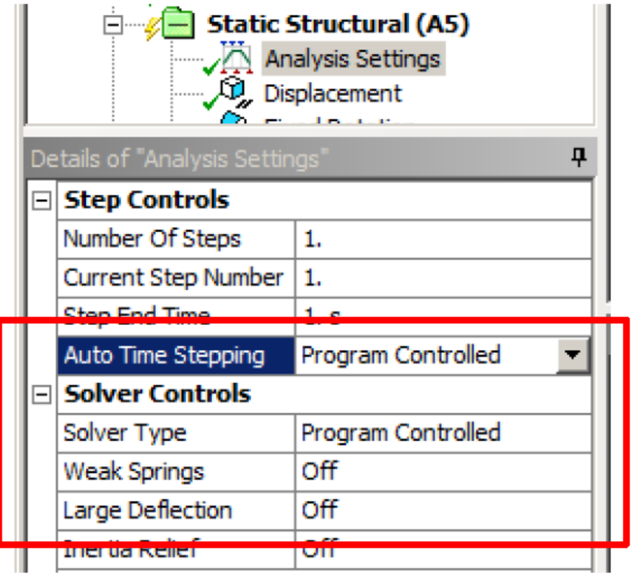
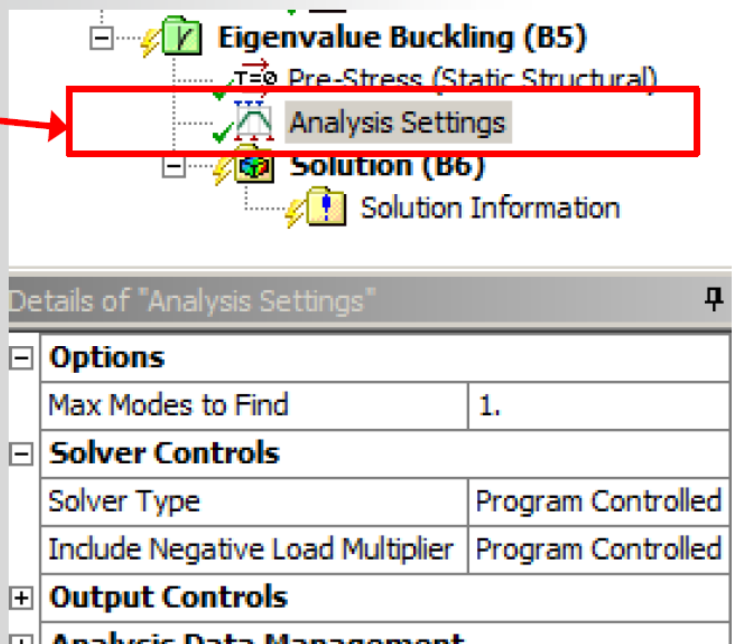
Note the Static Structural Analysis Settings Specifications:

This is a linear analysis

- **Auto Time Stepping = Program Controlled**
- **Large Deflection = Off**



For the Linear Buckling, we will take the default Analysis Settings to calculate the 1st eigenvalue load required to induce buckling.



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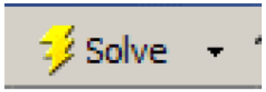
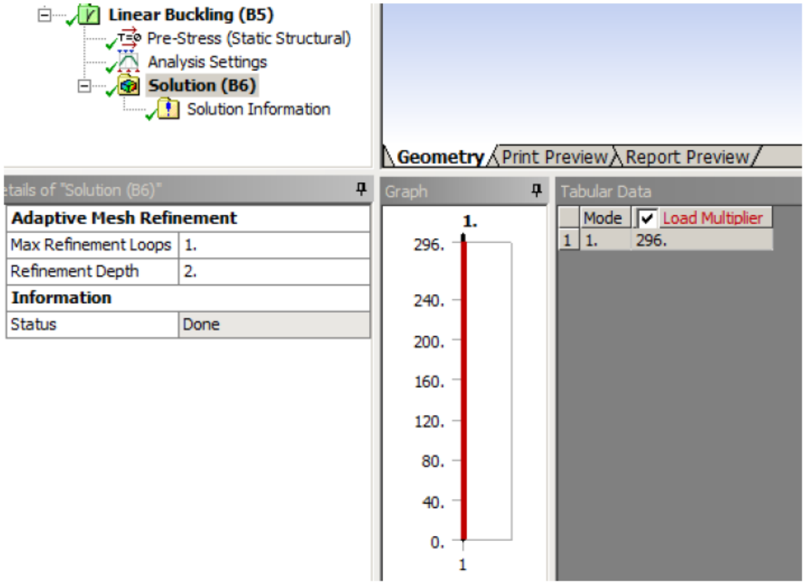
Run the prestressed Eigenvalue buckling solution:



Highlight the Solution Branch of the Eigenvalue Buckling Analysis to find the Load Multiplier.

Since a unit force load was applied, this load multiplier value (296) represents the magnitude of load necessary to begin a perturbation that leads to buckling.



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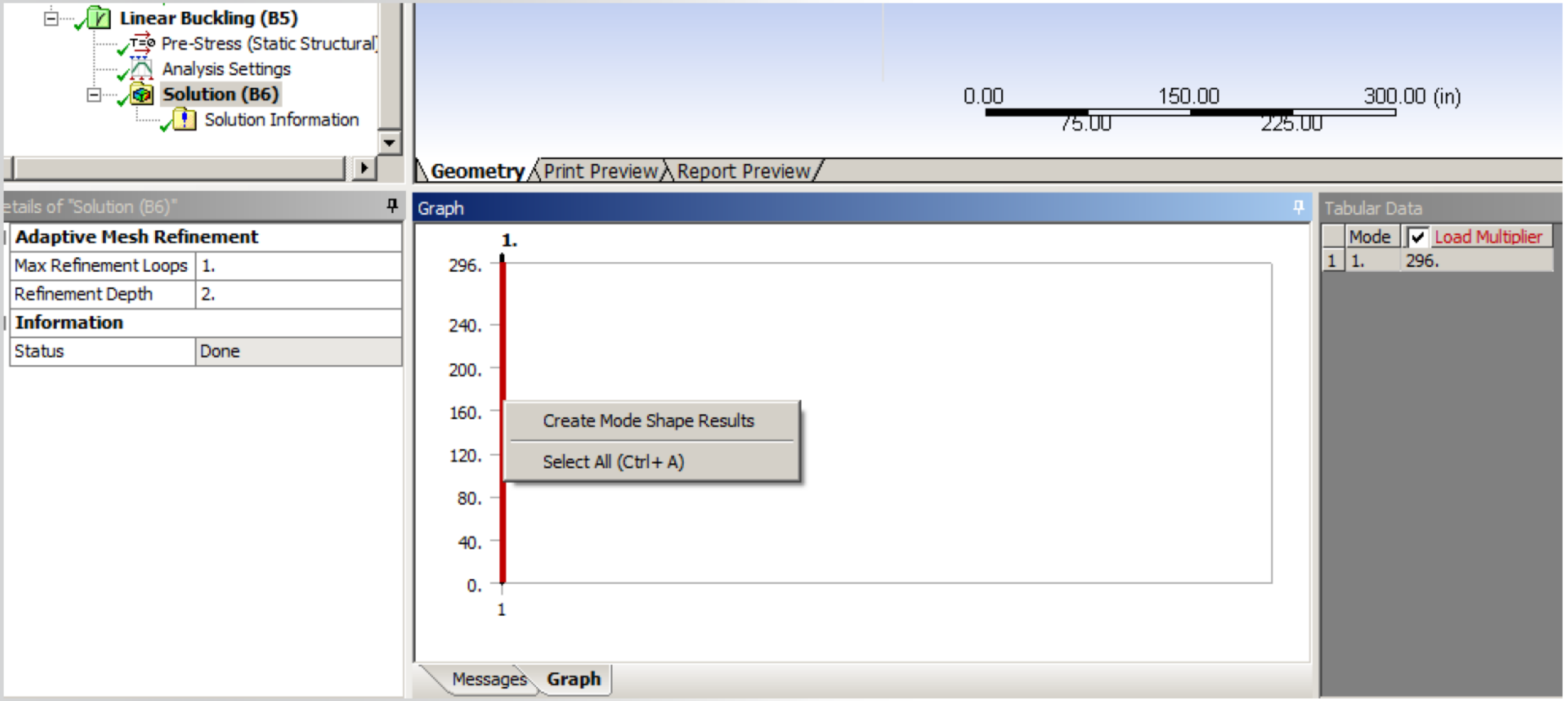
RMB on the buckling solution to "Create Mode Shape Results"



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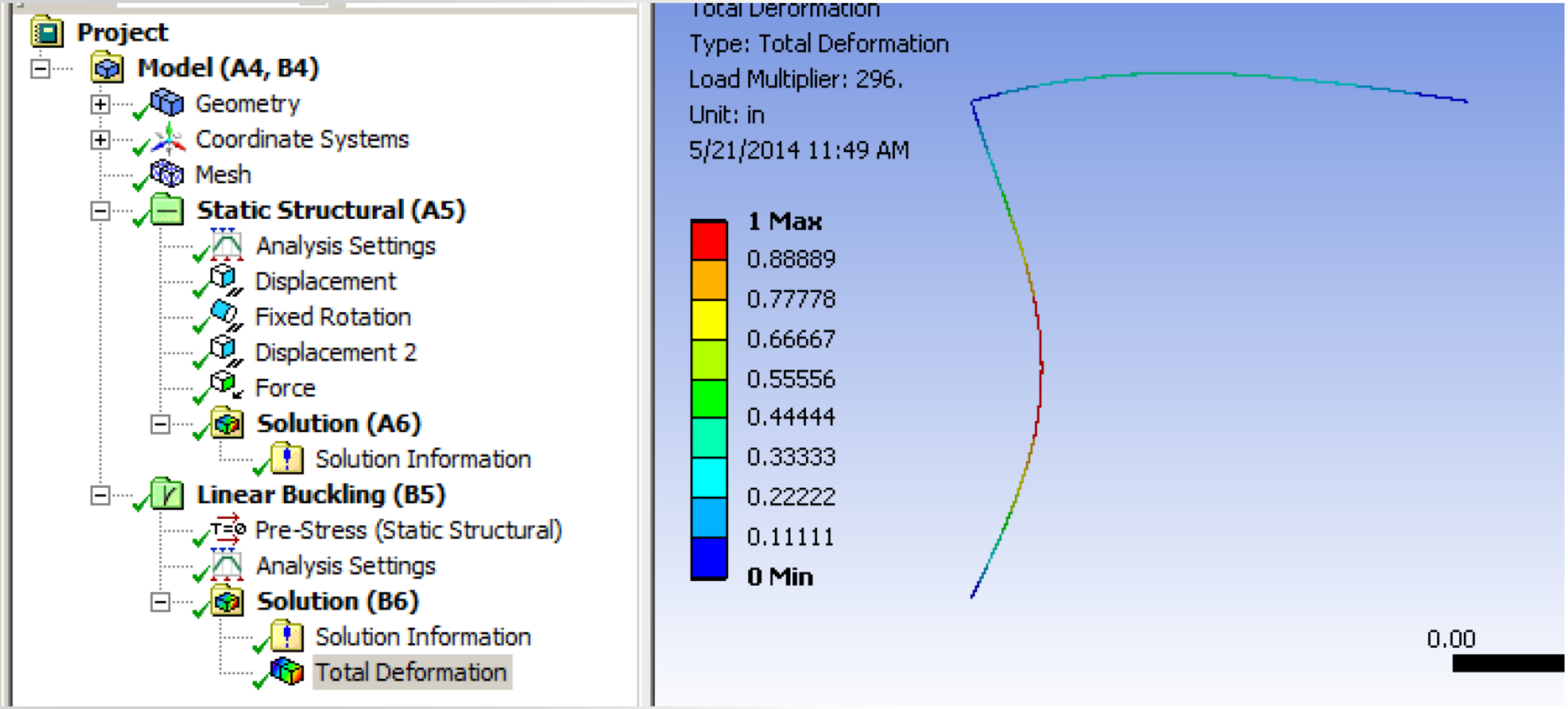
Evaluate Total Deformation results (Use results scaling to exaggerate the displacement).



Keep this Project open for the next workshop